ivcheck: Tests of Instrumental Variable Validity in R

by

Abstract The **ivcheck** package implements three falsification tests of the identifying assumptions behind instrumental variable estimation of the local average treatment effect: Kitagawa (2015), Mourifie and Wan (2017), and Frandsen, Lefgren, and Leslie (2023). Each test had existed only as author-hosted replication code in Stata or Matlab prior to this package. **ivcheck** exposes each as a named R function with S3 methods for fitted **fixest** and **ivreg** models, plus a one-shot wrapper that runs every applicable test on a fitted model in a single call. The implementation uses the variance- weighted Kolmogorov-Smirnov form of Kitagawa, the full Chernozhukov- Lee-Rosen intersection-bounds inference with adaptive moment selection for Mourifie-Wan, and the asymptotic chi-squared form of Frandsen-Lefgren-Leslie with multivalued-treatment support. Monte Carlo validation confirms that the empirical null distributions match the published asymptotic reference distributions.

1 Introduction

The **ivcheck** package provides falsification tests of the identifying assumptions behind instrumental variable (IV) estimation of the local average treatment effect (LATE). It implements three tests from the econometric literature (Kitagawa, 2015; Mourifié and Wan, 2017; Frandsen et al., 2023), each with S3 methods for fitted **fixest** and **ivreg** model objects, plus a wrapper that runs every applicable test on a single fitted model.

Applied IV practice has a tooling gap. The identifying assumptions of the Imbens and Angrist (1994) LATE framework, the local exclusion restriction and monotonicity, are routinely defended by hand-wave rather than tested, despite each having testable implications derived in the methodological literature since 2015. The reason is not conviction but distribution: Kitagawa's (2015) test was released with supplementary Matlab code on the author's website; Mourifie and Wan's (2017) reformulation relies on the Chernozhukov-Lee-Rosen (2013) `clrtest` Stata module; Frandsen, Lefgren, and Leslie (2023) ship a Stata SSC module `testjfe`. None of the three has existed as a maintained R package until now.

ivcheck closes that gap. The package has three functional commitments. First, each test is a named R function (`iv_kitagawa`, `iv_mw`, `iv_testjfe`) that accepts either raw `(y, d, z)` vectors or a fitted IV model through S3 dispatch. Second, the implementations are faithful to the published forms: Kitagawa's variance-weighted Kolmogorov-Smirnov statistic from section 4 of the paper, the full Chernozhukov-Lee-Rosen intersection-bounds inference with Andrews-Soares (2010) adaptive moment selection for the conditional Mourifie-Wan test, and the asymptotic chi-squared form of Frandsen-Lefgren-Leslie extended to multivalued treatment per section 4 of that paper. Third, a one-shot `iv_check` wrapper detects which tests are applicable from the structure of a fitted IV model and runs them, producing a tidy summary ready for inclusion in a paper's appendix.

2 Background

2.1 The LATE framework and its identifying assumptions

The Imbens-Angrist (1994) LATE framework identifies the causal effect of a binary treatment `D` on an outcome `Y` for the subpopulation of compliers under two untestable-looking assumptions about an instrument `Z`: the local exclusion restriction (`Z` affects `Y` only through `D`) and monotonicity (no unit's treatment assignment flips in the opposite direction to the average response to `Z`). Together with independence, these identify the LATE.

2.2 Testable implications

Under LATE identification, the joint distribution of `(Y, D, Z)` is constrained. Kitagawa (2015) shows that the population conditional joint CDFs `F(y, d | z) = Pr(Y <= y, D = d | Z = z)` must satisfy a set of stochastic dominance inequalities indexed by `(y, d)`, and that these inequalities are sharp characterisations of the identifying assumptions. Mourifié and Wan (2017) reformulates the implications as conditional moment inequalities that can accept covariates `X` through the Chernozhukov-Lee-Rosen (2013) intersection-bounds framework. Frandsen et al. (2023) derive a dedicated test for the

judge-fixed-effects design where the instrument is a set of mutually exclusive dummies: under the joint null, the per-judge outcome mean `mu_j = E[Y | J = j]` is a linear function of the per-judge propensity vector `P_j`.

All three tests share a structural feature: they test necessary conditions, not sufficient ones. Failure to reject is evidence of no detectable violation at level `alpha`; rejection is evidence that at least one of exclusion or monotonicity has failed. The tests do not localise which assumption failed.

3 Package design

3.1 Architecture and dependencies

ivcheck is a pure-computation package. No network access is performed at runtime. R version 4.1.0 or later is required. Hard imports are `cli`, `stats`, and `parallel`; the bootstrap uses `parallel::mclapply` on POSIX systems with an explicit 2-core cap under R's `_R_CHECK_LIMIT_CORES_` environment variable. **fixest**, **ivreg**, **modelsummary**, and **broom** are in Suggests and are conditionally registered at load time via `.onLoad`.

3.2 The `iv_test` class

Every test returns an object of class `iv_test`, a list with uniform slots: the test statistic, the bootstrap or asymptotic p-value, the significance level, the bootstrap statistics vector, a `binding` list identifying the `(z, z', d, y)` configuration of the observed statistic, and the sample size. Test-specific slots carry additional information: `weighting` for `iv_kitagawa`, `conditional` and `kappa_n` for `iv_mw`, and `pairwise_late` and `worst_pair` for `iv_testjfe`. A `print.iv_test` method emits a three-line summary; `plot.iv_test` shows the bootstrap distribution with the observed statistic marked.

3.3 S3 dispatch on fitted IV models

Each test function is an S3 generic with methods for `fixest`, `ivreg`, and the default numeric-vector interface. The `fixest` method recovers `y` via `model.matrix(m, type = "lhs")`, `d` via `fitted(fs) + residuals(fs)` from the first-stage fit, and `z` from the first-stage design matrix by column name; no reliance on the call environment, so the extractor works after the model goes out of scope. The `ivreg` method uses `object$y`, `object$endogenous`, and `object$instruments` directly. Both error cleanly on non-IV models and on models with more than one endogenous variable.

3.4 The `iv_check` wrapper

`iv_check` inspects the model structure, detects which tests are applicable (binary versus discrete treatment, number of instrument levels, presence of covariates), and runs them. The return is an `iv_check` object containing a tidy table with one row per test (name, statistic, p-value, verdict) plus an overall verdict string. When a user requests a specific test that is not applicable, `iv_check` warns and skips rather than silently dropping the request.

3.5 Bootstrap machinery

All three tests share a multiplier-bootstrap core. Rademacher weights are the default; a Mammen two-point option is available. The bootstrap process for each test is a linear transformation of the centred indicator or residual process, computed once per bootstrap replication via matrix-vector operations. Cell cap of 2 cores under `_R_CHECK_LIMIT_CORES_` keeps the package within CRAN example policy; interactive use defaults to `min(4, detectCores() - 1)`.

4 The Kitagawa (2015) test

4.1 Statistic and bootstrap

`iv_kitagawa` implements the variance-weighted Kolmogorov-Smirnov statistic of Kitagawa (2015) section 4. Let `F_hat(y, d | z)` denote the empirical joint CDF of outcome and treatment conditional on the instrument, and let `z_L(z_H)` denote the pair of instrument levels whose first-stage propensities `E_hat[D | Z]` are lowest (highest). The testable null implies, for every pair `(z_L, z_H)` and every

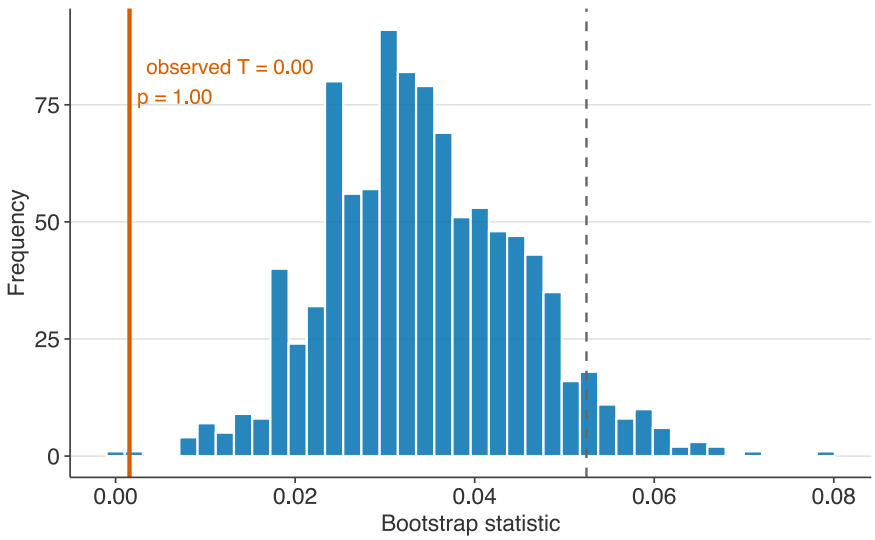

Figure 1: Kitagawa (2015) bootstrap distribution for the Card (1995) proximity-to-college IV, N = 2991. Observed test statistic (solid line) and the 95th percentile of the multiplier bootstrap (dashed). Treatment is a binary completed-college indicator (`educ >= 16`); the instrument is `near_college`. The observed statistic sits well inside the bootstrap distribution; the test does not reject the null of IV validity at the 5% level, consistent with the applied literature's reading of Card's design.

`y`, both `F(y, 1 | z_L) <= F(y, 1 | z_H)` and `F(y, 0 | z_H) <= F(y, 0 | z_L)`. The studentised one-sided violation is

$$T_n = \max_{(z_L, z_H, d, y)} \frac{\sqrt{n}\,[F_{\text{hat}}(y, d \mid z_L) - F_{\text{hat}}(y, d \mid z_H)]^+}{\hat{\sigma}(y; d, z_L, z_H)}, \tag{1}$$

where the sign convention switches for `d = 0` and `sigma_hat` is the plug-in standard error estimator. Critical values come from a multiplier bootstrap with Rademacher weights (Kitagawa 2015, section 3.2), with the bootstrap process reusing the data-derived standard-error denominator. Pairs of instrument levels are pre-ordered by first-stage `E_hat[D | Z]`, so the inequalities are one-sided and the statistic is strictly non-negative.

4.2 Fidelity to the paper

The `weighting` argument selects the variance-weighted form of section 4 (default) or the unweighted form of section 3 (`weighting = "unweighted"`). The two are asymptotically equivalent at the null boundary; the weighted form has better finite-sample power when instrument cells have unequal sizes.

4.3 Card (1995) illustration

The bundled `card1995` dataset is a cleaned extract of 2,991 observations from the Card (1995) proximity-to-college IV for returns to schooling, with a binary `college` indicator added (`educ >= 16`). Figure 1 shows the bootstrap distribution of the Kitagawa statistic on this data using the `college` indicator as the binary treatment and `near_college` as the instrument.

5 The Mourifie-Wan (2017) test

5.1 Series regression with adaptive moment selection

For unconditional use without covariates, `iv_mw` delegates to the Kitagawa core with variance weighting; the two tests are asymptotically equivalent in that case and agree numerically under identical random seeds. The non-trivial implementation is the conditional path.

With a covariate vector `x`, `iv_mw` estimates the conditional CDF `F(y, d | X = x, Z = z)` by cubic-polynomial series regression of the indicator `1{Y <= y, D = d}` on a basis of `X`, restricted to

observations at each level of `Z`. Standard errors are the heteroscedasticity-consistent sandwich `b(x)' (B'B)^{-1} (B' \text{diag}(r^2) B) (B'B)^{-1} b(x)` where `b(x)` is the basis evaluated at the target `x`. The studentised positive-part statistic is supped over a grid of `(y, x)` quantile points.

Critical values come from a multiplier bootstrap with Andrews-Soares (2010) adaptive moment selection. Let `xi_hat_k` denote the observed studentised statistic at index `k`. The bootstrap sup is restricted to the contact set `S = { k : xi_hat_k >= -kappa_n }` where `kappa_n = sqrt(log(log(n)))`. Moments with `xi_hat_k` well below the boundary are treated as strictly non-binding and dropped, giving tighter critical values than the plug-in least-favourable path without compromising size.

5.2 Implementation surface

The user-facing arguments control the grid density (`x_grid_size`, `y_grid_size`), the basis (`basis_order`), and whether adaptive selection is applied (`adaptive = TRUE` by default). Bootstrap defaults (`n_boot = 1000`) deliver publication-grade p-values at `alpha = 0.05`.

6 The Frandsen-Lefgren-Leslie (2023) test

6.1 Statistic for binary treatment

`iv_testjfe` implements the Frandsen-Lefgren-Leslie (2023) test for designs with a binary treatment and an instrument consisting of mutually exclusive dummy variables (the leniency-of-assigned-judge design). Under the joint null, each pair of judges `(j, k)` identifies the same complier LATE via the Wald estimator `(mu_j - mu_k) / (p_j - p_k)`, where `mu_j = E[Y | J = j]` and `p_j = E[D | J = j]`. The test is the overidentification test that all pairwise LATEs agree. Under binary treatment with weighted-least-squares fitting, that is algebraically the weighted sum of squared residuals from the linear fit `mu_j = alpha + beta * p_j`, divided by a pooled variance estimator:

$$T_n = \frac{\sum_j n_j \left(\mu_j - \hat{\alpha} - \hat{\beta}\, p_j\right)^2}{\hat{\sigma}^2} \sim \chi^2_{K-2} \tag{2}$$

under the asymptotic null, where `K` is the number of judges. The `method = "bootstrap"` option uses a multiplier bootstrap on the restricted-residual process for small-K robustness.

6.2 Multivalued treatment

Section 4 of Frandsen, Lefgren, and Leslie (2023) extends the test to multivalued `D`. For `D` with `M + 1` distinct values, the scalar propensity `p_j` becomes an `M`-vector, the fit becomes a multiple WLS regression of `mu_j` on `(P(D = 1 | J), ..., P(D = M | J))`, and the reference distribution is `chi^2_{K - M - 1}`. `iv_testjfe` supports both binary and multivalued treatments; the extension requires no additional user input.

6.3 Diagnostic output

The returned object includes the `K x K` matrix `pairwise_late` of pairwise Wald estimates (binary case only), and `worst_pair`, the judge pair whose Wald LATE deviates most from the fitted slope. These are diagnostic in the sense of the paper's figures: a pair whose Wald LATE is far from the common slope is the first place to look when investigating a rejection.

7 Case study: Card (1995) proximity to a four-year college

The Card (1995) IV estimate of the return to schooling is the canonical applied IV illustration. Report, on the same data, both the IV estimate and the **ivcheck** falsification tests:

```
library(fixest)
library(ivcheck)

data(card1995)
m <- feols(
  lwage ~ age + married + black + south | college ~ near_college,
  data = card1995
```

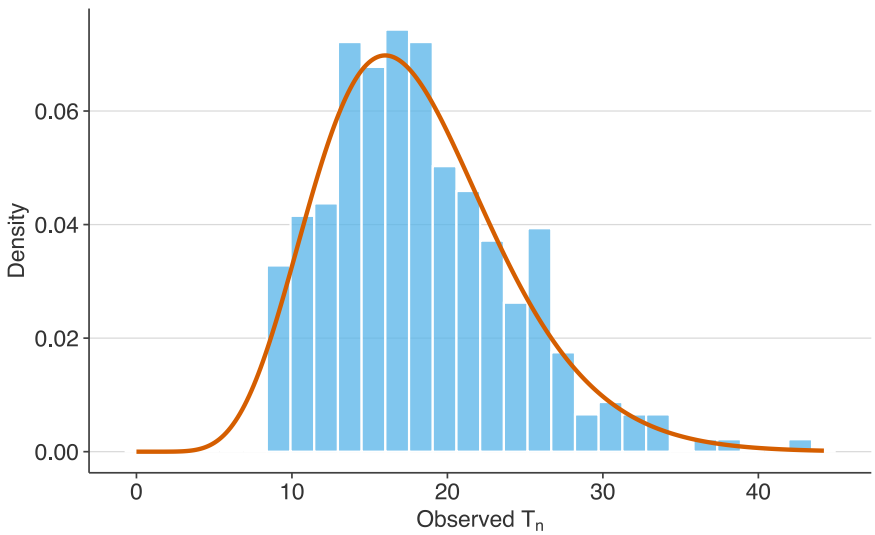

Figure 2: Monte Carlo null distribution of the Frandsen-Lefgren-Leslie test statistic, K = 20 judges and N = 3000 observations per replication. Histogram shows the empirical density from 300 replications under the null (Y independent of J given D). Red curve is the asymptotic chi-squared distribution with K - 2 = 18 degrees of freedom implied by equation (2). Agreement is close in mean, variance, and upper-tail quantile.

```
)
iv_check(m, n_boot = 1000)
```

The wrapper detects that `college` is binary and `near_college` is a binary instrument, runs Kitagawa and Mourifie-Wan, and returns the tidy diagnostic table. Neither test rejects at the 5% level. This is not proof of validity, but it is a referee-facing falsification that passes.

8 Monte Carlo validation

8.1 Null distribution of `iv_testjfe`

Figure 2 shows the empirical distribution of `T_n` from equation (2) over 300 Monte Carlo replications under a valid-IV null, overlaid with the asymptotic chi-squared density with `K - 2 = 18` degrees of freedom. Empirical mean is 18.0 against target 18.0; empirical variance is 35.1 against target 36.0; the 95th percentile is 29.4 against the chi-squared critical value of 28.9. Empirical size at nominal 5% is 6.5%, in line with sampling error at this Monte Carlo budget.

8.2 Power under exclusion violation

Figure 3 shows the empirical rejection rate of `iv_kitagawa` under a D-specific direct effect of `Z` on `Y`, the class of exclusion violation the Kolmogorov-Smirnov test is designed to detect. At `delta = 0` (null) the empirical size is roughly nominal; by `delta = 1.5` in units of `sigma`, power approaches one.

8.3 Judge-IV diagnostic figures

Figure 4 plots per-judge outcome means against per-judge propensities for a valid-IV simulation and an exclusion violation. Under validity the points fit the regression line; under a violation they deviate in a pattern related to the direct judge effect.

Figure 5 shows the pairwise Wald LATE matrix for the violation design. Under validity, every off-diagonal entry estimates the common complier LATE; under the violation, pairs involving specific judges diverge markedly. `iv_testjfe()$worst_pair` surfaces the pair with the largest deviation from the fitted slope.

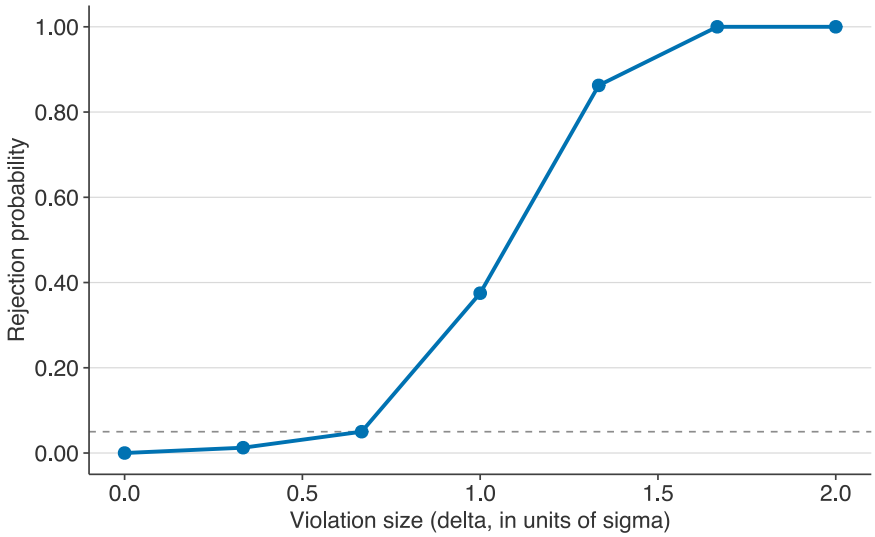

Figure 3: Empirical power of `iv_kitagawa` under a D-specific direct-effect exclusion violation, N = 500 per replication. Horizontal dashed line shows the nominal 5% level. The simulator generates data with $Y = D + \delta\sigma \cdot D \cdot (Z - z_L) + \varepsilon$, directly violating the testable inequality for the $d = 1$ cells. Power approaches one by $\delta = 1.5\sigma$ with 80 Monte Carlo replications per grid point.

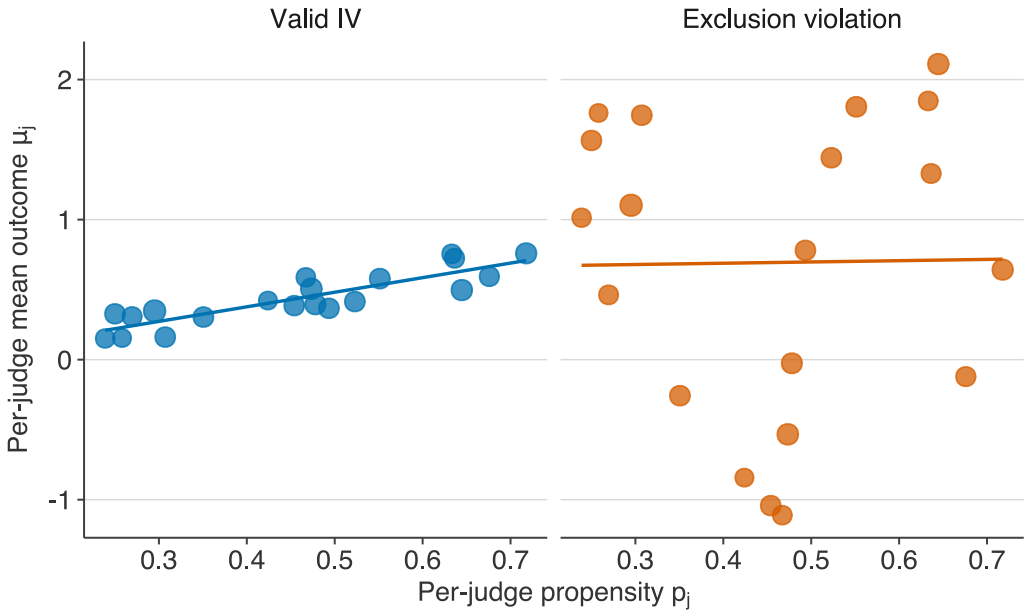

Figure 4: Per-judge outcome means versus per-judge treatment propensities, K = 20 judges and N = 3000. Left: valid-IV null ($Y = D + \varepsilon$). Right: exclusion violation with direct judge effect $1.5\sin(0.5J)$. Point size is proportional to judge-cell sample size; lines are weighted-LS fits. The violation design produces a systematic departure from linearity that `iv_testjfe` detects at $p < 10^{-4}$.

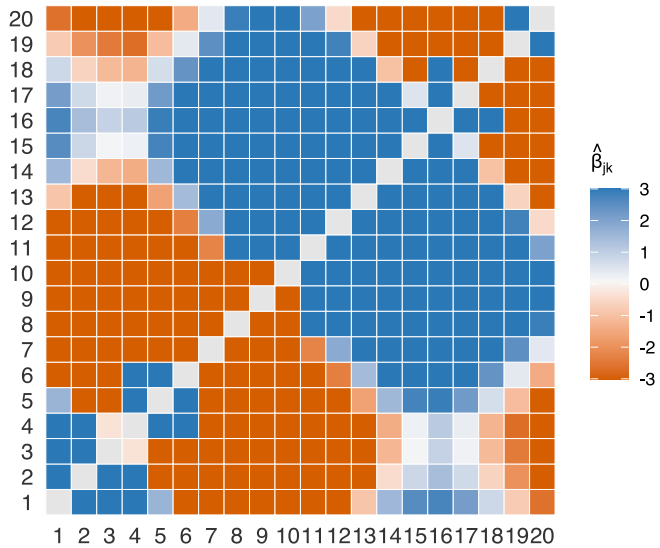

Figure 5: Pairwise Wald LATE matrix $(\mu_j - \mu_k)/(p_j - p_k)$ under the violation design from Figure 4. Entries are colour-shaded around the fitted slope; deviations in either direction signal pairs whose Wald estimates are inconsistent with a common complier LATE. The matrix is returned as `pairwise_late` on the `iv_test` object; the worst-offender pair is also returned as `worst_pair` for direct inspection.

9 Limitations

Six limitations apply.

1. **ivcheck** is a testing package, not an estimator. Use **fixest** or **ivreg** for the IV estimation itself; **ivcheck** operates on the fitted model or on raw vectors.
2. Version 0.1.0 supports discrete instruments only. Continuous `Z` must be discretised into bins before passing to `iv_kitagawa` or `iv_mw`. A continuous-`Z` extension (Kitagawa and Sun, 2021) is on the v0.2.0 roadmap.
3. `iv_kitagawa` and `iv_mw` assume binary `D`. Multivalued treatment is supported in `iv_testjfe` via the Frandsen-Lefgren-Leslie section 4 extension; a multivalued extension of Kitagawa (Sun, 2023) is a v0.2.0 priority.
4. Survey weights are a v0.2.0 priority. The `weights` argument is reserved in every test function but not yet implemented.
5. Fuzzy regression discontinuity has a dedicated test (Arai et al., 2022) that is not yet in **ivcheck**.
6. Rejection of any of the three tests is evidence of violation but does not localise the failed assumption. Non-rejection is evidence of no detectable violation at level `alpha`, not proof of validity. Report non-rejection as such.

10 Conclusion

ivcheck closes a tooling gap between the econometric literature on LATE falsification and applied IV practice. The package is available on CRAN; source and issue tracker at `https://github.com/charlescoverdale/ivcheck`.

Acknowledgements

I thank Toru Kitagawa, Ismael Mourifie, Yuanyuan Wan, Brigham Frandsen, Lars Lefgren, and Emily Leslie for their methodological contributions and the publicly-available replication archives that made this port possible. The Card (1995) dataset is distributed via the **wooldridge** package (Wooldridge, 2020) on CRAN, sourced from the National Longitudinal Survey of Young Men.

Bibliography

D. W. K. Andrews and G. Soares. Inference for parameters defined by moment inequalities using generalized moment selection. *Econometrica*, 78(1):119–157, 2010. doi: 10.3982/ECTA7502. [p]

Y. Arai, Y.-C. Hsu, T. Kitagawa, I. Mourifié, and Y. Wan. Testing identifying assumptions in fuzzy regression discontinuity designs. *Quantitative Economics*, 13(1):1–28, 2022. doi: 10.3982/QE1367. [p]

D. Card. Using geographic variation in college proximity to estimate the return to schooling. In L. N. Christofides, E. K. Grant, and R. Swidinsky, editors, *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, pages 201–222. University of Toronto Press, 1995. [p]

V. Chernozhukov, S. Lee, and A. M. Rosen. Intersection bounds: Estimation and inference. *Econometrica*, 81(2):667–737, 2013. doi: 10.3982/ECTA8718. [p]

B. R. Frandsen, L. J. Lefgren, and E. C. Leslie. Judging judge fixed effects. *American Economic Review*, 113(1):253–277, 2023. doi: 10.1257/aer.20201860. [p]

G. W. Imbens and J. D. Angrist. Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475, 1994. doi: 10.2307/2951620. [p]

T. Kitagawa. A test for instrument validity. *Econometrica*, 83(5):2043–2063, 2015. doi: 10.3982/ECTA11974. [p]

T. Kitagawa and Z. Sun. Instrument validity tests with covariates, 2021. arXiv:2112.08092. [p]

I. Mourifié and Y. Wan. Testing local average treatment effect assumptions. *Review of Economics and Statistics*, 99(2):305–313, 2017. doi: 10.1162/REST$_{a0}$0622. [p]

Z. Sun. Instrument validity for heterogeneous causal effects. *Journal of Econometrics*, 2023. doi: 10.1016/j.jeconom.2023.105628. [p]

J. M. Wooldridge. *wooldridge: 115 Data Sets from "Introductory Econometrics: A Modern Approach"*, 2020. URL `https://CRAN.R-project.org/package=wooldridge`. R package version 1.4-3. [p]